

- 9 (a) (i) $\Delta N / \Delta t$ (ignore any sign) B1 [1]
- (ii) $\Delta N / N$ (ignore any sign) B1 [1]
- (b) source must decay by 8% C1
- $A = A_0 \exp(-\ln 2 \ t / T_{1/2})$ or $A / A_0 = 1 / (2^{t/T})$ C1
- $0.92 = \exp(-\ln 2 \times t / 5.27)$ or $0.92 = 1 / (2^{t/5.27})$ C1
- $t = 0.634$ years
- $= 230$ days A1 [4]
- (allow 2 marks for $A / A_0 = 0.08$, answer 7010 days
allow 1 mark for $A / A_0 = 0.12$, answer 5880 days)

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Section B

10 (c) (part of) the output is added to /returned to / mixed with the input

P1